

UNIT II

Speech Recognition



Front-end Processor

Pre-emphasis

Frame-blocking and Windowing

Feature Extraction

Front-end Processor

In a traditional speech recognition systems, the pipeline for front-end processor has the following steps:

1. Preemphasis
2. Frame blocking and windowing
3. Feature Extraction

Step #1 — Preemphasis

- Pre-emphasis boosts the amount of energy in the high frequencies.
- For voiced segments like vowels, there is more energy at the lower frequencies than the higher frequencies.
- This is called *spectral tilt* which is related to the glottal source (how vocal folds produce sound).
- Boosting the high-frequency energy makes information in higher formants more available to the acoustic model.
- Pre-emphasis is used to make the system less susceptible to noise introduced in the process later.

Step #1 — Preemphasis

- Pre-emphasis compensates for the typical spectral tilt of speech and improves the relative strength of higher-frequency components.
- It amplifies higher frequencies that naturally drop off during human speech production.
- A first-order filter is commonly used in the front end.
- The output of the preemphasis, $s'(n)$ of the signal $s(n)$, where $\alpha \approx 0.95$ to 0.98 , is

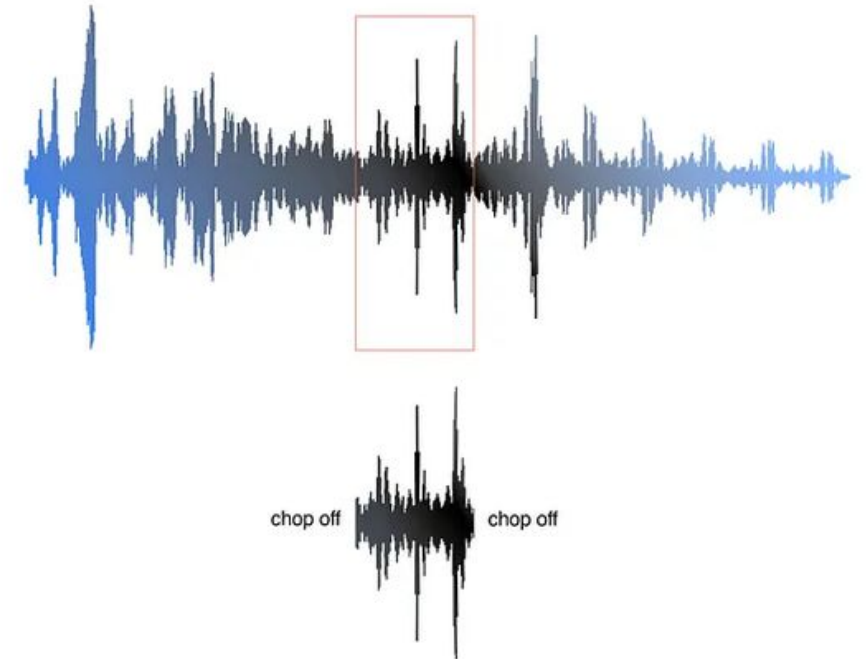
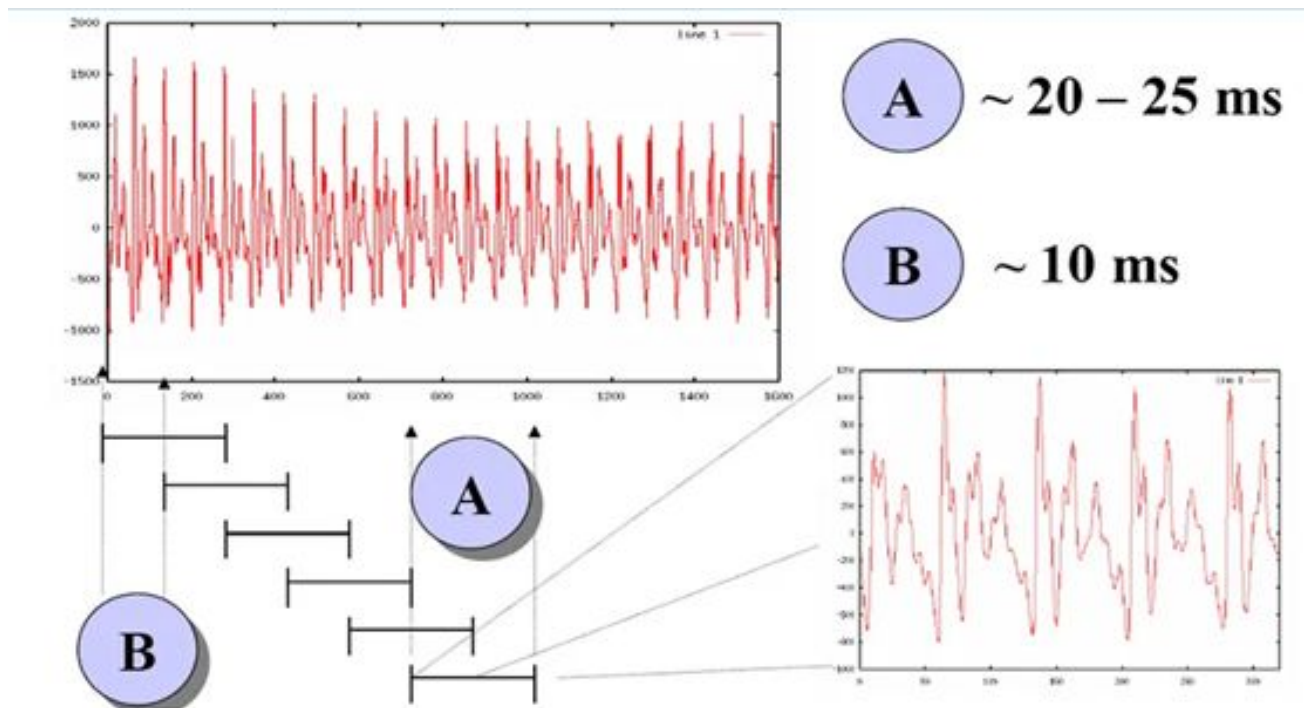
Typical form: $s'(n) = s(n) - \alpha s(n-1)$

Step #2 - Frame blocking and Windowing

- This step involves the slicing of the audio waveform into sliding frames.
- It cannot be just chop it off at the edge of the frame.
- The suddenly fallen in amplitude will create a lot of noise that shows up in the high-frequency.
- To slice the audio, the amplitude should gradually drop off near the edge of a frame.
- **Frame blocking:** segment the speech waveform into short overlapping frames so that the signal can be treated as approximately stationary.
- **Windowing:** Windowing multiplies each isolated frame by a mathematical curve (such as a Hamming or Hann window) that tapers the edges down smoothly to zero while keeping the center intact.

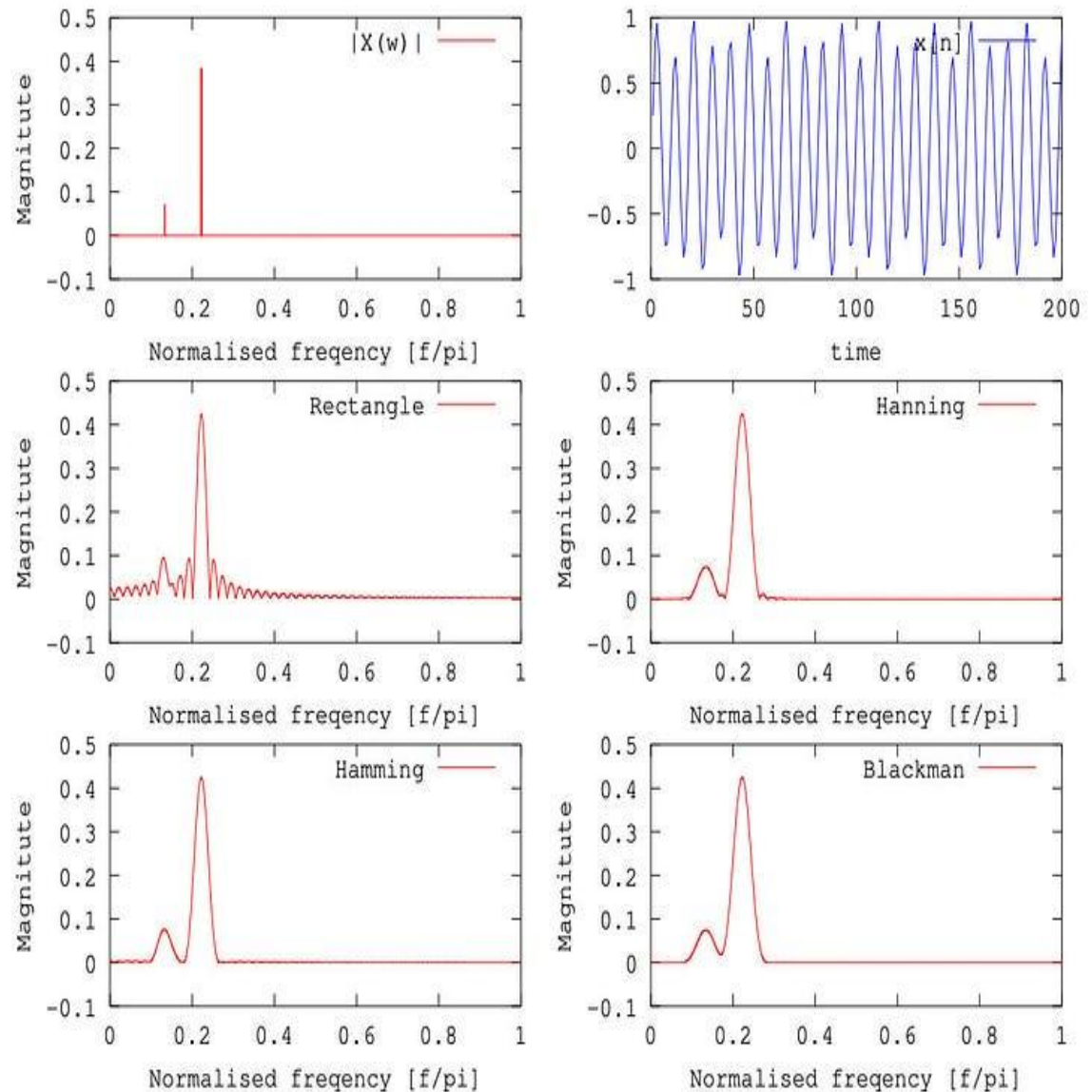
Step #2 - Frame blocking and Windowing

- n is the segmentation length and m is the hop applied to the original audio clip in the time domain.
- Window may be the Hamming window and the Hanning window.



Step #2 - Frame blocking and Windowing

- The chopped frame with Hamming and Hanning maintains the original frequency information better with less noise compared to a rectangle window.



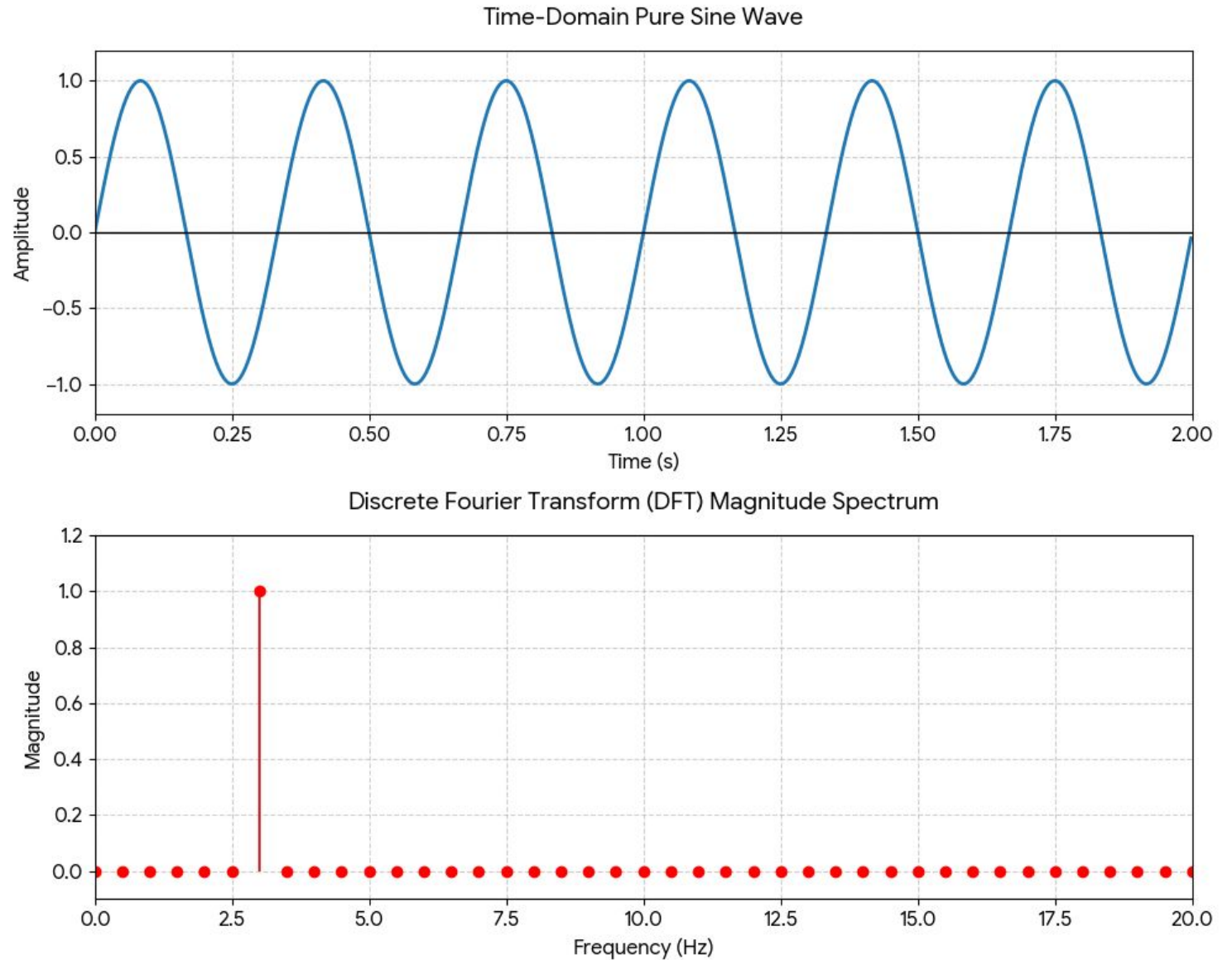
Step #4 — Feature Extraction

- A **speech feature** is a measurable data point extracted from an audio signal that isolates a specific, meaningful attribute for analysis.
- Speech features describe the **psychological and physiological components** that make up a single sound (phoneme).
- **Feature extraction** is the process of converting a speech signal into a compact, robust sequence of numerical vectors.
- Speech waveforms are highly variable, contain immense noise, and have massive data rates. Feature extraction condenses the signal into a **parametric representation** with a minimized data rate.
- It discards irrelevant background noise and speaker variability while preserving the critical phonetic or acoustic information.

Feature Extraction

Example:

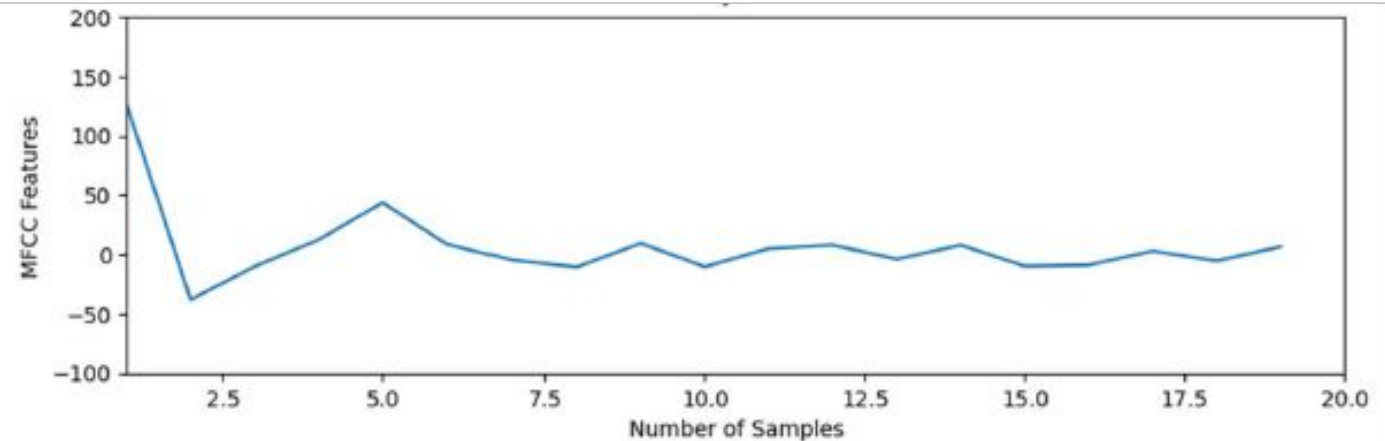
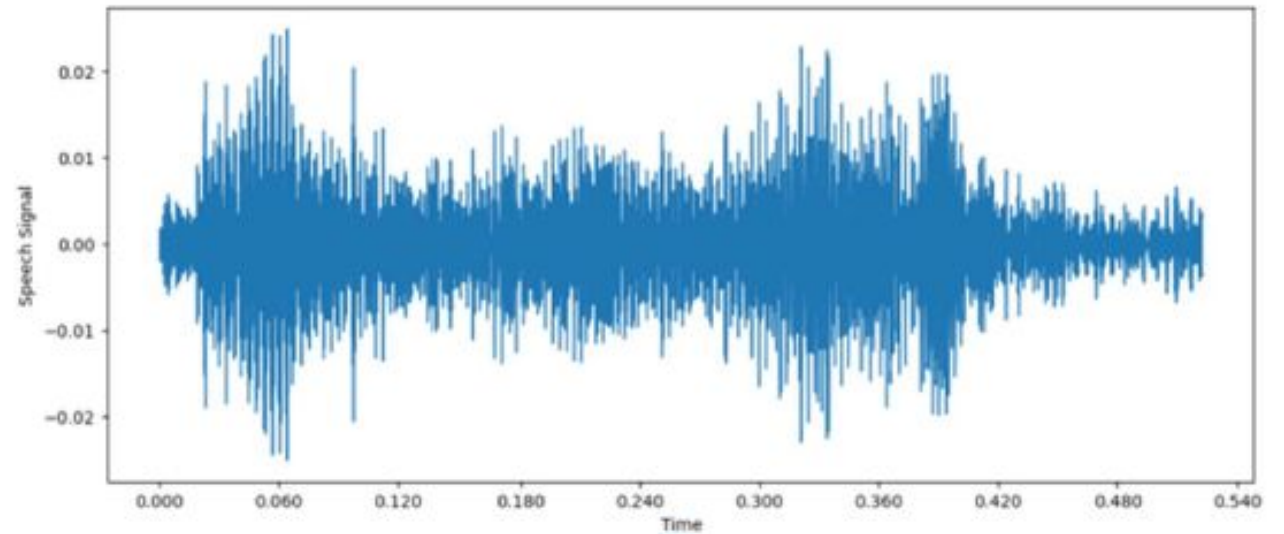
- Using DFT -
from a sine wave



Feature Extraction

Example:

- Using MFCC -
from a speech signal





Features

Types of Features

Robustness of Features

Feature Selection

Static vs Dynamic Features

- Static Features:

It is characteristics extracted from a segment of data that describe its overall properties over a fixed window of time, ignoring how the signal changes moment to moment.

- Dynamic Features :

Dynamic features capture how the acoustic properties of a voice change over time.

Dynamic features track the movement, speed, and acceleration of the speech articulators (tongue, lips, jaw) across successive frames.

First and second derivatives of features to track how the sound changes over time.

Types of Static Features

Types of static features are

- Temporal (Time-Domain) Features:
 - Root Mean Square (RMS) Energy, Zero-Crossing Rate
- Spectral (Frequency-Domain) Features:
 - Spectrogram, MFCC, LPCC
- Prosodic / Long-Term Features:
 - Pitch, Rhythm, intonation, and stress patterns (utterance-level).
- Glottal Features
 - Jitter, Harmonics-to-Noise Ratio
- Deep Learning & Learned Embeddings:
 - Convolution Neural Network (CNN).

Temporal Features

Temporal (Time-Domain) Features are features extracted directly from the raw waveform over time.

Examples:

1. **Root Mean Square (RMS) Energy:** Measures amplitude or loudness variations across frames.
2. **Zero-Crossing Rate (ZCR):** Tracks how often a signal changes sign (+ to -).

It is used for distinguishing voiced speech (low ZCR, high energy) from unvoiced speech / silence (high ZCR, low energy).

Spectral Features

Spectral (Frequency-Domain) Features are features derived by transforming the signal into the frequency domain using Short-Time Fourier Transform (STFT) or Filter banks.

Examples:

1. **Spectrogram:** A visual map plotting frequency changing over time, mapping signal intensity.
2. **Spectral Centroid:** A shape descriptor of Power Spectrum. Pinpoints the "center of mass" of the spectrum; correlates to sound "brightness".
3. **Spectral Bandwidth:** Measures the frequency spread around the centroid.

Cepstral Features

Cepstral features are features warped mathematically to mimic how the human ear actually filters and perceives sound scales.

Examples:

1. MFCC:

Mel-Frequency Cepstral Coefficients (MFCCs) are specialized features that mimic how human ears process frequencies that are used for speech recognition software.

2. LPCC:

Cepstral coefficients derived from the Linear Predictive Coding (LPC) spectral envelope. It models the vocal tract as a digital filter, representing the resonant frequencies (formants) of speech.

Prosodic Features

Spans across multiple phone segments, syllables, or entire utterances, carrying emotional and stylistic context.

Examples:

- 1. Fundamental Frequency (F_0 / Pitch):** The rate of vocal cord vibration; reflects intonation and stress.
- 2. Duration / Rhythm:** Speaking rate, pause durations, and vowel elongation.
- 3. Intensity:** Variations in volume over long segments.

Used in Speech Emotion Recognition (SER), emphasis detection, and text-to-speech synthesis (TTS).

Glottal Features

Measures the micro-variations and irregularities in the speaker's vocal fold (glottis) vibration.

Examples:

1. **Jitter:** Short-term variations in the pitch (F_0) period (frequency instability).
 2. **Shimmer:** Short-term variations in the peak-to-peak amplitude (loudness instability).
 3. **Harmonics-to-Noise Ratio (HNR):** Ratio of periodic energy to additive noise in the voice.
- Used in Clinical speech pathology (detecting voice disorders), age/gender classification, and speaker identification.

Deep Learning Features/Representations

High-level latent representations extracted automatically via deep neural networks rather than explicit mathematical formulas.

1. **Spectrogram Representations:** Deep neural networks (CNNs) treat 2D spectrograms as images.
2. **Self-Supervised Embeddings:** Dense vector representations extracted from massive pre-trained models (e.g., wav2vec 2.0, HuBERT, Whisper embeddings).
3. **Speaker Embeddings:** Vectors designed specifically to isolate speaker identity (e.g., d-vectors, x-vectors).

Dynamic Features

Examples of Dynamic Features of Speech Signal:

1. Delta (or Velocity):

- First-order temporal derivatives of the static features.
- Measure the rate of change of a specific feature from one frame to the next.
- The delta shows the trajectory of how a sound is morphing into the next sound (e.g., transitioning from an "ah" to a "ee")

2. Delta-Delta (or Acceleration):

- Second-order temporal derivatives, calculated by taking the derivative of the Delta features.
- It tells the model whether the rate of change is speeding up or slowing down.

Robustness of Features

Robustness of features refers to how well a specific feature representation maintains its core information and stability when exposed to noise, distortions, shifts in the environment, or adversarial tampering.

Common Challenges to Audio Robustness:

- **Additive Noise:** Unwanted sounds mixed directly into the audio signal, such as street noise, wind, music, or babbling crowds.
- **Convolutional (Channel) Distortion:** Changes introduced by the hardware itself—such as the frequency response of a simple smartphone microphone, a laptop mic, or a VoIP telephone line.
- **Reverberation:** Sound waves bouncing off walls and objects, smearing the audio signal across time and muddying distinct acoustic transitions.

Robustness of Features

Some techniques to improve robustness of features.

Mean and Variance Normalization: Because channel distortion (like a low-quality microphone) acts as a constant multiplier on the frequency spectrum, it shifts baseline feature values. This process normalizes the features over an utterance (or sliding window) to have a mean of zero and a variance of one. This effectively strips away constant channel biases.

Dynamic Features (Deltas and Delta-Deltas): As discussed previously, static spectral features can be heavily warped by steady background noise. By incorporating temporal derivatives (velocities and accelerations), the model focuses on how frequencies transition relative to one another over time rather than their absolute energy levels.

Robustness of Features

Multi-Condition Data Augmentation: Training models on clean audio artificially corrupted with thousands of varieties of background noise, impulse responses, and pitch/speed shifts so the network learns to ignore them natively.

Front-End Denoising Networks: Using deep learning architectures (such as U-Nets or regression DNNs) as a preprocessing front-end to clean up log-power spectra or filter spectrograms before downstream feature extraction and classification take place.

Feature Selection

In speech processing, feature selection (and feature reduction) is the process of identifying and keeping only the most informative parts of your acoustic data while discarding redundant features.

For example, adjacent frequency bands in a spectrogram or adjacent time frames in a delta stream often carry overlapping information.

Passing all features straight into a model can lead to **overfitting, high computational costs, and confusion from redundant signals.**

Feature Selection

Common feature selection techniques in speech processing are

1. Statistical Filter Methods:

These evaluate features independently of the model using metrics like Mutual Information or correlation. For example, in traditional speech analysis, coefficients with very low variance or high correlation to noise are dropped early.

2. Dimensionality Reduction (Transformative Selection): Transformation techniques map the full feature set into a smaller, highly optimized space.

PCA (Principal Component Analysis): Keeps dimensions that capture the highest variance.

LDA (Linear Discriminant Analysis): Selects and projects features based on what actually helps separate classes (like phonemes or speakers), making it a favorite for traditional speech pipelines.